

Highlights

Literature-Grounded Research Ideation with State-of-the-Art Pipelines and Facet-Grounded Gap Verification

Paulo Roberto de Souza, Frederico Araújo Durão

- Explicit SOTA tier definition with dual-pass retrieval and hybrid re-ranking over open scholarly catalogs.
- Reproducible landscape $\rightarrow$ gap matrix $\rightarrow$ proposal pipeline with evidence-only generation rules.
- FGGV extension for facet-level novelty, gap alignment, and diversity-aware recommendation lists.
- Formal notation, equations, algorithms, and online evaluation protocol for transparent auditing.

Literature-Grounded Research Ideation with State-of-the-Art Pipelines and Facet-Grounded Gap Verification

Paulo Roberto de Souza^{a,*}, Frederico Araújo Durão^a

^a*Institute of Computing, Federal University of Bahia, Adhemar de Barros
Avenue, Salvador, 40170-110, Bahia, Brazil*

Abstract

Many computational techniques have been proposed to help researchers explore scientific directions, yet the volume of published literature often exceeds what any individual can survey before proposing new work. This problem, known as *literature overload*, makes it difficult to filter relevant papers, identify open gaps, and generate ideas that are genuinely grounded in evidence. This paper presents POST-REC (Paper-Oriented Scientific Topic Recommender), a reproducible ideation system that models research proposal generation as a recommender problem over literature-backed items. The proposed method combines dual-pass bibliographic retrieval over OpenAlex, deterministic relevance filtering, hybrid BM25–dense–vector re-ranking with explicit state-of-the-art (SOTA) tier quotas, and a multi-stage pipeline that materializes landscape summaries and gap matrices before idea generation. An extension called *Facet-Grounded Gap Verification* (FGGV) adds literature facet maps, gap–facet alignment, contrastive facet novelty indices, and diversity-aware selection. We formalize notation, component metrics, composite ranking equations, and pseudocode algorithms. A motivation scenario in computer science illustrates relevance scoring, gap-matrix construction, and facet-level verification. The deployed system evaluates methods online through a blind A/B study between SOTA and FGGV, collecting volunteer feedback via the Expectation Alignment Score and related metrics.

*Corresponding author.

Email addresses: paulo.prsdesouza@ufba.br (Paulo Roberto de Souza),
fdurao@ufba.br (Frederico Araújo Durão)

Keywords: Recommender systems, research ideation, state of the art, retrieval-augmented generation, facet novelty, expert systems

1. Introduction

The exponential growth of scholarly publishing has increased the difficulty of identifying relevant literature, recognizing underserved research gaps, and proposing novel directions that remain faithful to current evidence [1, 2]. A computer-science researcher investigating retrieval-augmented generation (RAG) for literature-grounded research ideation, for example, may need to analyze thousands of papers on LLM agents, hybrid retrieval, and novelty verification before articulating a feasible proposal. Too much information can quickly exceed cognitive limits, while unconstrained large language model (LLM) outputs may invent citations, recycle saturated ideas, or hide how “state of the art” was defined [3].

Recommender systems (RSs) have emerged as a practical response to overload problems by filtering items according to user preferences [4, 1]. In POST-REC, the recommended *items* are structured research ideas; user preferences are expressed through seed topics, research area, depth, constraints, and subsequent feedback. Retrieval-augmented generation (RAG) [5] grounds generation in retrieved documents, but RAG alone does not specify SOTA tiers, explicit gap matrices, or facet-level verification. Recent ideation agents—including ResearchAgent [6], NOVA [7], SciMON [8], Scideator [9], and the Idea Novelty Checker [10]—improve retrieval and novelty assessment, yet rarely expose all methodological artifacts within one reproducible pipeline.

This paper aims to investigate, model, and quantify literature-grounded research ideation by proposing a transparent pipeline that:

1. defines SOTA through configurable recency and citation tiers;
2. constructs an explicit gap matrix before proposal generation;
3. verifies novelty at document and facet levels;
4. learns from volunteer feedback through an Expectation Alignment Score (EAS).

The main contributions of this work are:

- **The SOTA-grounded ideation pipeline**, which produces landscape summaries and a gap matrix before generating proposals linked to concrete evidence papers;

- **The hybrid retrieval and relevance model**, combining dual-pass OpenAlex fetching, lexical relevance scoring, BM25 + dense embedding + pgvector ranking, and SOTA tier quotas;
- **Facet-Grounded Gap Verification (FGGV)**, extending the baseline with literature facet maps (LFMs), gap–facet alignment (GFA), contrastive facet novelty (CFN), and facet diversity selection (FDS);
- **An online evaluation protocol**, centered on blind A/B assignment between SOTA and FGGV in Auto mode, with volunteer feedback aggregated on a validation dashboard.

The remaining aspects of this paper are structured as follows. Section 2 discusses related work. Section 3 introduces the proposed system, including notation, equations, algorithms, and a motivation scenario. Section 4 presents the experimental evaluation. Section 5 discusses limitations, and Section 6 concludes the paper and presents future work.

2. Related work

In the realm of literature-grounded ideation, numerous studies employ retrieval, planning, or facet-based verification to recommend research directions. However, a significant gap remains in jointly exposing explicit SOTA tiering, gap-matrix artifacts, and facet-level verification within one pipeline.

Table 1 compares representative systems. Si et al. [2] provide rigorous human evaluation but no facet map or gap matrix. ResearchAgent [6] and NOVA [7] emphasize iterative expansion over graphs or planned search at higher computational cost. Scideator [9] and the Idea Novelty Checker [10] apply facet-level checks, typically post-hoc rather than during gap-driven generation. SciMON [8] optimizes contrastive novelty against inspiration papers but does not materialize a landscape matrix artifact. Chain of Ideas [11] models temporal literature chains without an explicit gap cell structure.

Unlike these approaches, POST-REC materializes `sota_landscape` and `gap_matrix` JSON artifacts before proposal generation, making gap-driven ideation inspectable by reviewers. Hybrid lexical–neural retrieval follows established practice in scientific search [12], while facet re-ranking relates to RankGPT-style rerankers [13].

Table 1: Comparison of literature-grounded ideation approaches (2023–2026).

System	Retrieval	Gap structure	Novelty unit	Limitation
Si et al.	Semantic Scholar API	Implicit in abstracts	Human-rated originality	No facet map
ResearchAgent	Citation graph	Agent review loops	Multi-agent debate	High cost
NOVA	Planned embedding search	Iterative broadening	Planning steps	Weak SOTA tiering
Scideator	Paper-pool facets	UI recombination	Facet-level checks	Weak gap alignment
INC	Retrieve–rerank	Post-hoc	Per-facet LLM compare	Not gap-driven
POST-Rec SOTA	OpenAlex dual-pass + hybrid rank	Landscape + gap matrix	Document + verified signals	LLM-synthesized gaps
POST-Rec FGGV	Same as SOTA	Gap matrix + LFM	Per-facet FNI + GFA	Online sample size

3. The proposal

This section describes the proposed POST-REC system. It begins with a recommendation overview. Subsequently, it presents the bibliographic data features, notation, document preprocessing, component metrics, composite ranking equations, and recommendation algorithms. A motivation scenario with numeric illustration is provided to demonstrate how the methodology operates in practice.

3.1. Recommendation overview

Figure 1 summarizes the macro pipeline. Given a user expectation E , the system executes five major steps: (1) expectation parsing and topic expansion; (2) dual-pass retrieval over OpenAlex [14]; (3) relevance filtering and hybrid re-ranking; (4) mode-dependent generation (Quick single-pass or

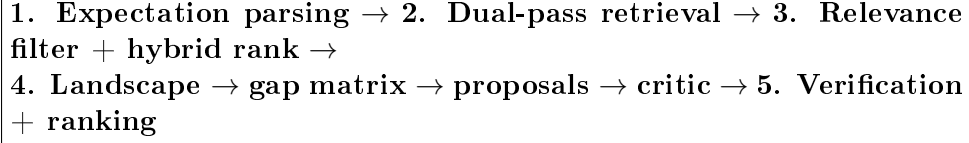


Figure 1: Overview of the POST-REC recommendation pipeline.

SOTA/FGGV multi-stage); and (5) verification and final ranking. Progress events in the deployed interface map one-to-one to these stages.

3.2. Bibliographic data features

POST-REC uses OpenAlex as its primary literature source, with optional normalization from arXiv, Semantic Scholar, and Crossref metadata. Each document is represented by features used across retrieval, ranking, and generation:

- **Text features:** title, abstract, and venue for lexical overlap and embedding;
- **Bibliographic features:** DOI, publication year, and citation count for deduplication and tiering;
- **Domain features:** field, subfield, and topic tags for alignment filtering.

Only papers that pass relevance filtering and hybrid ranking are passed to the generation stage.

3.3. Notation

Understanding the notation is crucial for comprehending the equations and algorithms presented below. Table 2 summarizes the main symbols.

The expanded topic set is defined as

$$T^* = \text{unique}(T_{\text{seed}} \cup T_{\text{learned}} \cup T_{\text{techniques}}), \quad (1)$$

and query tokens are $Q = \text{tokens}(T^* \cup A)$ after stopword removal and token-length filtering ($|\text{token}| > 2$). The retained paper set is

$$\mathcal{P} = \{p \in \mathcal{D} \mid s_{\text{rel}}(p) \geq \tau_{\text{rel}}\}, \quad (2)$$

with default threshold $\tau_{\text{rel}} = 0.22$.

Table 2: Notation used in the proposed method.

Symbol	Description
U	Researcher user
E	User expectation $\langle T_{\text{seed}}, A, O, D, C \rangle$
T_{seed}	Seed topic phrases
A	Research area
Q	Query token set for retrieval and relevance
$p \in \mathcal{D}$	Normalized paper
$O(X, Q)$	Token overlap ratio $ X \cap Q / Q $
$s_{\text{rel}}(p)$	Paper relevance score in $[0, 1]$
e_p	Embedding of paper p
$\mathcal{S}_{\text{SOTA}}$	SOTA-recent tier subset
$\mathcal{S}_{\text{found}}$	Foundation (seminal) tier subset
G	Gap matrix from landscape analysis
FNI_f	Facet Novelty Index for facet f
GFA	Gap-facet alignment score
S_{verified}	Verified final ranking score

3.4. Document preprocessing and relevance filtering

Retrieved papers are normalized (title, DOI, year, source), deduplicated, and scored before any LLM call. For each paper p , token sets $T_{\text{title}}(p)$ and $T_{\text{body}}(p)$ are built from the title and from title||abstract||venue, respectively. The overlap function is

$$O(X, Q) = \frac{|X \cap Q|}{|Q|}, \quad Q \neq \emptyset. \quad (3)$$

The relevance score is

$$s_{\text{rel}}(p) = w_t O(T_{\text{title}}, Q) + w_b O(T_{\text{body}}, Q) + b_{\text{cite}} - p_{\text{avoid}}, \quad (4)$$

$$b_{\text{cite}} = \min\left(\frac{\ln(1 + c_p)}{\delta}, b_{\text{max}}\right), \quad (5)$$

with defaults $w_t = 0.55$, $w_b = 0.30$, $\delta = 12$, $b_{\text{max}} = 0.15$, and avoided-topic penalty $p_{\text{avoid}} = 0.35$ when applicable. If both title and body overlap are very low, the score is capped at 0.18.

3.5. SOTA tier metrics

SOTA membership is defined explicitly rather than implicitly through retrieval alone:

$$p \in \mathcal{S}_{\text{SOTA}} \Leftrightarrow \text{year}(p) \geq y_{\text{now}} - Y, \quad (6)$$

$$p \in \mathcal{S}_{\text{found}} \Leftrightarrow c_p \geq \tau_c, \quad (7)$$

with defaults $Y = 3$ and $\tau_c = 50$. Dual-pass retrieval queries OpenAlex separately for recent SOTA and foundation tiers, then merges the results.

3.6. Hybrid re-ranking metric

Each $p \in \mathcal{P}$ receives a sparse score $s_{\text{BM25}}(p, Q)$ and a dense score $s_{\text{dense}}(p, Q) = \cos(e_p, e_Q)$. The hybrid score blends both signals (sparse weight 0.4 by default) with pgvector retrieval (blend factor 0.25). After sorting, tier quota $\phi = 0.6$ ensures that fraction of the top- k context papers belong to $\mathcal{S}_{\text{SOTA}}$.

3.7. Landscape and gap matrix

Let $\mathcal{P}_k$ denote the top- k papers after hybrid ranking. The SOTA pipeline:

1. clusters $\mathcal{P}_k$ into thematic groups;
2. summarizes problems, methods, datasets, evaluation protocols, and limitations per cluster;
3. builds gap matrix G with rows as sub-problems and columns as method/data/evaluation families;
4. labels each cell $G_{ij} \in \{\text{well-covered, partial, underserved}\}$.

The generator samples underserved cells and requires each proposal to cite SOTA anchors from $\mathcal{P}_k$. A critic pass rejects ungrounded or near-duplicate ideas.

3.8. Verified ranking and FGGV metrics

Each idea i receives LLM-estimated dimensions $\text{dim}_{i,j} \in [0, 100]$ and verified signals $\text{SOTA_fit}(i)$, $\text{novelty_verified}(i)$. The verified score is

$$S_{\text{verified}}(i) = \sum_j w_j \cdot \text{dim}_{i,j} + w_{\text{sota}} \cdot \text{SOTA_fit}(i) + w_{\text{nov}} \cdot \text{novelty_verified}(i). \quad (8)$$

FGGV extends Eq. (8) with facet-level structure. For facets $f \in \{\text{problem, method, data, evaluation}\}$

$$\text{FNI}_f(i) = 1 - \max_{s \in \text{LFM}_f} \text{sim}(\text{facet}_f(i), s), \quad (9)$$

and the composite FGGV score is

$$\text{FGGV}(i) = \alpha S_{\text{verified}}(i) + \beta \text{FNI}(i) + \gamma \text{GFA}(i) + \eta \cdot \text{doc_novelty}(i), \quad (10)$$

with defaults $\alpha = 0.30$, $\beta = 0.32$, $\gamma = 0.23$, $\eta = 0.15$.

3.9. Motivation scenario

To demonstrate the utility of the proposed method, this subsection presents a practical example of the full recommendation process, including relevance filtering, tier selection, gap-matrix construction, and facet scoring.

Consider a researcher U who submits:

- $T_{\text{seed}} = \{\text{retrieval-augmented generation, research ideation agents}\};$
- $A = \text{artificial intelligence};$
- $\text{mode} = \text{SOTA}.$

Paper p_1 (retained): title “*Benchmarking retrieval-augmented generation pipelines for scientific question answering*”, year 2024, citations 92. Token overlap yields $O(T_{\text{title}}, Q) = 0.45$ and $O(T_{\text{body}}, Q) = 0.31$. With $b_{\text{cite}} = 0.15$,

$$s_{\text{rel}}(p_1) = 0.55(0.45) + 0.30(0.31) + 0.15 = 0.490 > \tau_{\text{rel}}.$$

Paper p_2 (filtered): title “*Introduction to classical information retrieval models*”, year 1999, citations 680. Although $b_{\text{cite}} = 0.15$, weak overlap caps $s_{\text{rel}}(p_2) \leq 0.18$. Thus p_2 is excluded despite high citations.

After hybrid ranking, suppose $|\mathcal{P}| = 52$, with 31 `sota_recent` and 21 `seminal` papers. With context size 30 and quota $\phi = 0.6$, at least 18 SOTA-recent papers enter the generation context. A simplified gap matrix for this run is:

Table 3: Excerpt of gap matrix G for the motivation scenario.

	Hybrid retrieval	Facet verification	Human eval.
Literature-grounded ideation	partial	underserved	partial
Scientific QA with RAG	well-covered	partial	underserved

An generated idea targeting literature-grounded ideation $\times$ facet verification receives high GFA if it declares alignment with the underserved cell. If its method facet is distant from dominant single-pass RAG statements in the literature facet map, $\text{FNI}_{\text{method}}$ remains high (e.g., 0.76), supporting ranking under FGGV mode.

Algorithm 1 Paper relevance filtering

Require: Papers $\mathcal{D}$, query tokens Q , threshold τ_{rel}

Ensure: Filtered set $\mathcal{P}$

```
1:  $\mathcal{P} \leftarrow \emptyset$ 
2: for each  $p \in \mathcal{D}$  do
3:   Compute  $O(T_{\text{title}}, Q)$  and  $O(T_{\text{body}}, Q)$  using Eq. (3)
4:    $b_{\text{cite}} \leftarrow \text{Eq. (5)}$ 
5:    $s_{\text{rel}}(p) \leftarrow \text{Eq. (4)}$ ; apply weak-overlap cap
6:   if  $s_{\text{rel}}(p) \geq \tau_{\text{rel}}$  then
7:      $\mathcal{P} \leftarrow \mathcal{P} \cup \{p\}$ 
8:   end if
9: end for
10: return  $\mathcal{P}$ 
```

3.10. Recommendation algorithms

Algorithm 1 filters papers by relevance. Algorithm 2 performs SOTA-grounded generation. Algorithm 3 updates the user profile from feedback.

4. Experimental evaluation

This section presents the evaluation of the proposed system. In the current deployment, validation is performed **online**: researchers execute runs in the live platform, rate generated ideas, and contribute feedback aggregated by experimental variant. Supplementary offline scripts support regression testing during development but are not the principal evidence source.

4.1. Online evaluation setup

When a user selects **Auto** mode, the system assigns sticky buckets:

- **Control (SOTA)**: landscape $\rightarrow$ gap matrix $\rightarrow$ proposals $\rightarrow$ critic;
- **Treatment (FGGV)**: SOTA pipeline plus LFM, GFA, CFN, and FDS.

Assignment uses a deterministic hash of user identity and experiment identifier `fggv_vs_sota_v1`, with treatment fraction 0.5. Method labels and FGGV sub-scores are hidden during blind runs.

Algorithm 2 SOTA-grounded idea generation

Require: Expectation E , ranked papers $\mathcal{P}_k$, mode m

Ensure: Ranked idea set $\mathcal{I}$

- 1: Build landscape summary from $\mathcal{P}_k$
 - 2: Construct gap matrix G
 - 3: Sample underserved cells from G
 - 4: $\mathcal{I}_0 \leftarrow \text{LLM-GENERATE}(E, \mathcal{P}_k, G)$ under evidence-only rules
 - 5: $\mathcal{I}_1 \leftarrow \text{CRITIC}(\mathcal{I}_0, \mathcal{P}_k)$
 - 6: **if** $m = \text{fggv}$ **then**
 - 7: Build LFM; compute FNI and GFA; apply FDS
 - 8: Score ideas with Eq. (10)
 - 9: **else**
 - 10: Score ideas with Eq. (8)
 - 11: **end if**
 - 12: **return** $\mathcal{I}$ sorted by final score
-

Algorithm 3 Expectation Alignment Score and profile update

Require: Ratings: usefulness U , relevance R , clarity C , feasibility F , trust T , would-use W

Ensure: EAS and updated topic profile

- 1: Map $W \in \{\text{yes}, \text{maybe}, \text{no}\}$ to $\{5, 3, 1\}$
 - 2: $\text{EAS} \leftarrow 0.25U + 0.20R + 0.20C + 0.15F + 0.10T + 0.10W$
 - 3: Update T_{learned} or avoided topics
 - 4: **return** EAS
-

4.2. Evaluation metrics

The primary online metrics are:

- **Expectation Alignment Score (EAS)**

$$\text{EAS} = 0.25U + 0.20R + 0.20C + 0.15F + 0.10T + 0.10W; \quad (11)$$

- **Originality** — user-reported novelty of each idea;
- **Approval rate** — proportion of ideas marked as suitable for use in a paper or project.

Feedback updates T_{learned} and avoided topics, influencing subsequent retrieval through Q and p_{avoid} in Eq. (4). The validation dashboard aggregates means by `experiment_variant`.

4.3. Results and monitoring

In the current pilot, run counts, mean EAS, originality, and approval rates are monitored separately for `sota` and `fggv`. When sufficient ratings are available, significance tests on alignment differences are reported in the research-insights interface. Thus, the system supports continuous on-line comparison under realistic usage rather than relying solely on a fixed offline benchmark.

4.4. Supplementary offline checks

The repository provides `scripts/run_offline_evaluation.py` to replay golden topics and compare Quick (B1), SOTA (B2), and FGGV (M) with ablation variants. These checks verify implementation consistency when retrieval or scoring code changes; they do not replace volunteer judgments in the online study.

5. Discussion

The proposed methodology makes SOTA definition, gap structure, and ranking components explicit through notation, equations, and stored artifacts. This transparency supports auditing by reviewers and collaborators. The motivation scenario in Section 3.9 shows how lexical relevance, tier quotas, and gap-matrix alignment interact in a concrete computer-science run.

Current limitations include dependence on OpenAlex metadata quality, LLM-synthesized gap matrices, imperfect facet extraction, and the need for larger online samples before claiming superiority over all systems in Table 1. Online evaluation also depends on volunteer availability and self-selected topics, which may introduce sampling bias.

6. Conclusion

This paper presented POST-REC, a literature-grounded research ideation methodology combining hybrid retrieval, explicit SOTA tiers, gap-matrix-driven generation, verified ranking, and the FGGV extension for facet-level novelty and alignment. The proposal was described through notation, component metrics, composite equations, pseudocode algorithms, and a worked scenario. Online blind A/B evaluation between SOTA and FGGV provides the primary empirical validation path in the deployed system.

Future work includes deeper citation-graph integration, cross-system benchmarks against ResearchAgent and Scideator, multilingual catalog expansion, and scaling the online evaluation cohort. The open implementation aims to support reproducible comparison as the scientific ideation landscape continues to evolve.

Acknowledgements

This work was partially supported by CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brazil), Finance Code 001.

References

- [1] F. Ricci, L. Rokach, B. Shapira, Recommender systems handbook (2015).
- [2] C. Si, Y. Yang, T. Hashimoto, Can LLMs generate novel research ideas? A large-scale human study with 100+ NLP researchers, in: International Conference on Learning Representations, 2025.
- [3] C. Lu, C. Lu, R. Lange, et al., The AI scientist: Towards fully automated open-ended scientific discovery, arXiv preprint arXiv:2408.06292 (2024).
- [4] G. Adomavicius, A. Tuzhilin, Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions, IEEE Transactions on Knowledge and Data Engineering 17 (6) (2005) 734–749.
- [5] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks, Advances in Neural Information Processing Systems 33 (2020) 9459–9474.
- [6] J. Baek, A. F. Aji, A. Saffari, Researchagent: Iterative research idea generation over scientific literature with large language models, in: North American Chapter of the Association for Computational Linguistics, 2025.
- [7] Z. Hu, Y. Lu, W. Chen, H. Yu, Y. Zhang, W. Chen, Improving language models via plug-and-play retrieval feedback, in: Findings of the Association for Computational Linguistics, 2025.

- [8] Y. Wang, Y. Li, Z. Jiang, Z. Zhang, et al., Scimon: Scientific inspiration machines optimized for novelty, in: Association for Computational Linguistics, 2024.
- [9] S. Roy, et al., Scideator: Human-in-the-loop scientific idea generation, arXiv preprint arXiv:2404.00000 Accessed 2026 (2024).
- [10] K. Ramesh, et al., Idea novelty checker: Expert-labeled facet comparisons for scientific novelty assessment, in: SDP Workshop, ACL, 2025.
- [11] Y. Li, et al., Chain of ideas: Literature evolution chains for research ideation, Findings of EMNLP (2025).
- [12] G. Zuccon, B. Koopman, A. Deacon, et al., Integrating and evaluating neural and lexical retrieval models for health search, *Information Processing & Management* 51 (4) (2015) 475–488.
- [13] W. Sun, et al., Is chatgpt good at search? investigating large language models as re-ranking agents, arXiv preprint arXiv:2304.09542 (2023).
- [14] J. Priem, H. Piwowar, R. Orr, Openalex: A fully open index of scholarly works, authors, venues, and institutions, arXiv preprint arXiv:2205.01833 (2022).